Week of 23rd

Monday, 23 October 2023 5:14 PM

"Macro"

Control Volume Balance

$$0 = \frac{dm}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\, dS$$

$$\sum_{on\,CV} \underline{f} = \frac{d(m\underline{v})_{CV}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\,\underline{v}\, dS$$

→ Macro → Micro
- arbitrary CV
- Express terms as integrals
- Leibniz rule & divergence theorem

Microscopic Balance

$$\frac{\partial\rho}{\partial t} + \underline{v}\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

$$\rho\left(\frac{\partial\underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\tau} + \rho g$$

ρ = density
$\underline{v}$ = velocity
p = pressure
g = gravity

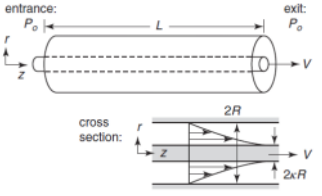
1) Choosing CV (can be arbitrary)
↓
writing out equations specific to that CV
↳ Have to rewrite & recalculate if CV is changed

→

1) Already written for an arbitrary, microscopic CV
↳ applicable to any problem ✓
↳ Just need to change boundary conditions & get velocity & pressure profiles!

PROBLEM: 6.49 Flow Problem: Axial annular drag with pressure drop, wire coating. Repeat Problem 43 with an imposed pressure gradient $-\Delta p/L$ in the flow direction. Calculate the velocity field only.

PROBLEM: 6.43 Flow Problem: Axial annular drag, wire coating. An incompressible Newtonian fluid fills the annular gap between a cylindrical wire of radius κR and an outer shell of inner radius R (Figure 6.22). The wire is pulled to the right at speed V. There is no pressure variation throughout the apparatus. Calculate the steady state velocity profile, the flow rate, and the force on the wire. This geometry occurs in wire-coating. Answer: [redacted]



→ Flow in z ✓
→ gradients in r ✓
→ Symmetric around θ ✓

In wire-coating, a wire is drawn through a bath. This axial drag flow is addressed in Problem 43.

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial\theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial\theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

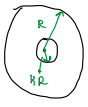
θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial\theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial\theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial\theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial\theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial\theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

6.43: No pressure variation ; $\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$; $\underline{g} = \begin{pmatrix} g_r \\ g_\theta \\ 0 \end{pmatrix}_{r\theta z}$



Mass balance / Continuity equation:

SS, Incompressible

$$\frac{\partial\rho}{\partial t} + v\cdot\nabla\rho = -\rho(\nabla\cdot\underline{v})$$

$$(\nabla\cdot\underline{v}) = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}(rv_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z} = 0$$

$$\boxed{\frac{\partial v_z}{\partial z} = 0}$$

Start w/ Navier stokes:

r-direction : $0 = \rho g_r$

θ-direction : $0 = \rho g_\theta$

θ- direction : $0 = \rho g_\theta$

z - direction : $\rho\left(v_z \frac{\partial v_z}{\partial z}\right) = \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{\partial^2 v_z}{\partial z^2}\right]$ (terms: $v_z\frac{\partial v_z}{\partial z}$ and $\frac{\partial^2 v_z}{\partial z^2}$ are 0)

$$\mu\left(\frac{1}{r}\right)\frac{\partial}{\partial r}\left[r\frac{\partial v_z}{\partial r}\right] = 0$$

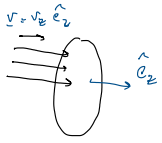
Solve for velocity profile

Boundary conditions: always have to be about the variable you need to solve in the PDE! (its value or its derivatives value)

@ $r = \kappa R$, $v_z = V$

@ $r = R$, $v_z = 0$

$$Q = \iint_S \hat{n}\cdot\underline{v}\, dS$$



$$= \int_0^{2\pi}\int_{\kappa R}^{R} v_z\, r\, dr\, d\theta$$

$$\hookrightarrow \int_{\kappa R}^{R} \underbrace{r}_{u}\underbrace{\ln r}_{dv}\cdot dr$$

easier to calculate derivative of !

Int. by parts: $\int u\,dv = uv - \int v\,du$ $\quad u = \ln r$ $\quad du = \frac{1}{r}\cdot dr$ $\quad v = \frac{r^2}{2}$ $\quad dv = r$

$$\int d(uv) = \int v\,du + \int u\,dv$$

$$uv = \int v\,du + \int u\,dv$$

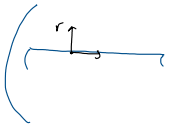
$$\int u\,dv = uv - \int v\,du$$

$$\int r\ln r = \frac{r^2}{2}\ln r - \int \frac{r}{2}\cdot dr$$

Force on wall?

$$F = \iint \left(\hat{n}\cdot\underline{\Pi}\right)\Big|_{@\, r=\kappa R} dS \qquad \Pi = -p\underline{I} + \underline{\tau}$$

$$=$$



$\hat{e}_r$

$$F = \iint \hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & -p & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix}_{r\theta z} (r)\, dr\, d\theta$$

κR Constant

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}_{r\theta z} = \mu \begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

(6.49) Pressure variation

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

HW problem

$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g_z \end{pmatrix}$$



r-direction

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}+v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r}+\rho g_r+\mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_r)\right)+\frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}+\frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r v_\theta}{r}+v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial \theta}+\rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv_\theta)\right)+\frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2}+\frac{2}{r^2}\frac{\partial v_r}{\partial \theta}+\frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z}+\rho g_z+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2}+\frac{\partial^2 v_z}{\partial z^2}\right]$$

Power Law : Non-newtonian behavior :

Start from Cauchy balance: $\rho\left(\frac{\partial \underline{v}}{\partial t}+\underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$

$$\underline{\underline{\tau}} = \eta(\dot\gamma)\,\underline{\underline{\dot\gamma}} = \eta(\dot\gamma)\left[\nabla\underline{v}+(\nabla\underline{v})^T\right] =$$

$$\mu\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} \\ r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta} & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_r}{\partial z}+\frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta}+\frac{\partial v_\theta}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}$$

where $\dot\gamma = |\underline{\underline{\dot\gamma}}|$

$$= +\sqrt{\frac{1}{2}\left(\begin{array}{c}\text{Sum of Squares}\\ \text{of each term}\end{array}\right)}$$

$$\eta(\dot\gamma) = m\,\dot\gamma^{n-1} \quad \Big\} \text{ Power law}$$

$$\therefore \tau = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$$

Momentum equation :

$$\rho\left(\frac{\partial \underline{v}}{\partial t}+\underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \nabla\cdot\underline{\underline{\tau}} + \rho\underline{g}$$

$$\begin{pmatrix} 0 \\ 0 \\ \cdot \end{pmatrix} \qquad \nabla\cdot\underline{\underline{\tau}} = \begin{pmatrix} \frac{\partial \tau_{zr}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial z}(r\tau_{rz}) \end{pmatrix}$$

$$\nabla\cdot\underline{\underline{A}} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}(rA_{rr}) + \frac{1}{r}\frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} - \frac{A_{\theta\theta}}{r} \\ \frac{1}{r}\frac{\partial}{\partial r}(r^2A_{r\theta}) + \frac{1}{r}\frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} \\ \frac{1}{r}\frac{\partial}{\partial r}(rA_{rz}) + \frac{1}{r}\frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}_{rtz} \quad \text{B.3-10}$$

$$\begin{pmatrix} 0 \\ 0 \\ \rho \end{pmatrix}_{z\text{-dir}} \qquad \nabla \cdot \overleftrightarrow{\tau} = \begin{pmatrix} \frac{\partial \tau_{zr}}{\partial z} \\ 0 \\ \frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rz}) \end{pmatrix}$$

$$\downarrow$$

$$\tau_{rz} = \tau_{zr} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) = \frac{(-1)}{(-1)}\, m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right)$$

$$= \frac{m}{(-1)}\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(-\frac{\partial v_z}{\partial r}\right)^{1}$$

$$\tau_{zr} = -m\left(-\frac{\partial v_z}{\partial r}\right)^{n}$$